

Dipayan Biswas

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Summary

Machine learning researcher and Ph.D. candidate specializing in multimodal AI: vision-language models, LLM evaluation, and computer vision. Deployed a summarization system used by 750+ students and published 5 peer-reviewed IEEE papers.

Education

University of Houston

Ph.D. in Computer Science, GPA 3.88

M.Sc. in Computer Science (awarded en route to the Ph.D.)

Houston, TX

Aug 2021 – Dec 2026 (Expected)

May 2026

Khulna University of Engineering & Technology (KUET)

B.Sc. in Electronics and Communication Engineering, GPA 3.88

Khulna, Bangladesh

Apr 2012 – May 2016

Technical Skills

AI/ML: Multimodal AI, Vision-Language Models (VLMs), Large Language Models (LLMs), LLM Evaluation (LLM-as-judge), Prompt Engineering, Fine-tuning (LoRA, QLoRA), Computer Vision, Object Detection

Frameworks: PyTorch, Hugging Face (Transformers, PEFT, TRL), OpenCV, LiteLLM, LangChain, DeepEval, scikit-learn

Models & APIs: Gemini, OpenAI, Anthropic, Gemma 3, Qwen3-VL, SigLIP, CLIP, YOLO, Faster R-CNN, DETR

Programming & MLOps: Python, SQL, C++, Git, FastAPI, vLLM, Ollama, MLflow, Slurm, AWS

Experience

University of Houston

Graduate Research / Teaching Assistant

Houston, TX

Aug 2021 – Present

Multimodal Video Summarization (NSF-funded VideoPoints platform)

- Condensed hour-long lectures into 5-minute summaries with supporting visuals for review and navigation, used by 750+ students in 10+ courses; 78% of surveyed students rated them helpful.
- Designed a multimodal pipeline (speech, text, slides), raising faculty-rated faithfulness by 19% and coverage by 15%.
- Automated evaluation with an LLM-as-judge that reproduced the faculty's 19% faithfulness gain (judge: 18%).
- Developed zero-shot cross-modal selection of supporting visuals, rated 82.8% by faculty (up from 70.3%).

Tools: Gemma 3, Qwen3-VL, Gemini API, Whisper, Tesseract OCR, SigLIP, vLLM, DeepEval, Slurm

Object Detection in Presentation Videos

- Released LVVO, a 4,000-frame dataset from 245 videos with multi-annotator expert labels (83.4% inter-annotator F1).
- Raised AP50 from 90.8% to 95.3% with semi-supervised auto-labeling of 3x more frames, without new manual labels.
- Developed a graph-based method merging fragmented detections into coherent visuals, improving mAP by 15.8%.

Tools: PyTorch, torchvision, Ultralytics YOLO, Faster R-CNN, DETR, OpenCV

Fellowship.AI

Machine Learning Fellow

Remote

Sep 2025 – Nov 2025

- Developed a multimodal analyzer scoring listing photos and text for pricing and staging across 3k real estate listings.
- Flagged mispriced properties at 85% precision with a weighted-KNN pipeline over 8K+ Redfin listings.

Tools: Python, Gemini API, scikit-learn, GeoPandas, Matplotlib, Firecrawl

Khulna University of Engineering & Technology (KUET)

Assistant Professor / Lecturer

Khulna, Bangladesh

Feb 2017 – Aug 2021

- Taught five undergraduate courses with labs, including Digital Image Processing, and supervised ML research projects.
- Mentored 10+ students, leading to IEEE papers on plant disease and breast cancer classification (99.8%, 99.3% accuracy).

Tools: Python, PyTorch, TensorFlow, scikit-learn, OpenCV, pandas

Samsung R&D Institute Bangladesh (SRBD)

Software Engineer

Dhaka, Bangladesh

Jul 2016 – Feb 2017

- Implemented 360-degree photo and video viewing modes, including little-planet, for the shipped Gear 360 iOS app.

Tools: Objective-C, iOS, OpenGL ES, OpenCV

Projects

- Multi-Step LLM and VLM Pipelines** [Code] *Jun 2026 – Sep 2026*
- Built configurable prompt chaining for multi-step workflows on local or cloud models, with resumable experiments.
- Unified Inference Toolkit for Vision-Language Models** [Code] *Jun 2026 – Aug 2026*
- Built one interface for multimodal prompts on cloud APIs, vLLM, Ollama, and Hugging Face Transformers.
- VLM Fine-tuning for Visual Question Answering** [Code] *May 2026 – Jun 2026*
- Fine-tuned Gemma 3 with QLoRA, raising ChartQA accuracy from 43.6% to 61.6% (4B) and 61.8% to 69.4% (12B).
- Lecture Video Chaptering with LLMs** [Details] *Jun 2025 – Aug 2025*
- Deployed automatic chaptering for every new VideoPoints lecture via the Gemini API; in daily use since Aug 2025.
- Modular Object Detection Training Pipeline** [Code] *Dec 2024 – Feb 2025*
- Designed a PyTorch codebase for reproducible YOLO, Faster R-CNN, and DETR training, with Hydra configs.
- Small and Distant Face Detection** [Code] *Feb 2023 – May 2023*
- Trained an IoU-aware RetinaFace on 4 V100 GPUs, improving AP on the medium and hard WIDER FACE sets.

Publications

- **D. Biswas**, S. Shah, and J. Subhlok. “Visual Content Detection in Educational Videos with Transfer Learning and Dataset Enrichment.” *IEEE MIPR 2025*. [Paper] [Code]
- **D. Biswas**, S. Shah, and J. Subhlok. “Lecture Video Visual Objects (LVVO) Dataset: A Benchmark for Visual Object Detection in Educational Videos.” *arXiv 2025*. [Paper] [Data]
- **D. Biswas**, S. Shah, and J. Subhlok. “Identification of Visual Objects in Lecture Videos with Color and Keypoints Analysis.” *IEEE ISM 2023*. [Paper]
- I. Z. Mukti and **D. Biswas**. “Transfer Learning Based Plant Diseases Detection Using ResNet50.” *IEEE EICT 2019*. [Paper]
- S. Das and **D. Biswas**. “Prediction of Breast Cancer Using Ensemble Learning.” *IEEE ICAEE 2019*. [Paper]
- **D. Biswas** and M. F. Hossain. “Epileptic Seizure Detection Based on Selected Features of Different Complexities Using ANN.” *IEEE EICT 2017*. [Paper]

Honors & Awards

- Ph.D. Research Showcase, Audience’s Choice Runner-Up, University of Houston, 2025
- Cullen Graduate Student Fellowship, University of Houston, 2021–2026
- University Gold Medalist (ranked 1st in class), KUET, 2018

Certifications & Courses

- Generative AI with Large Language Models, DeepLearning.AI and AWS [Certificate]
- Machine Learning in Production, DeepLearning.AI [Certificate]
- Building and Evaluating Advanced RAG, DeepLearning.AI [Course]
- AWS Cloud Practitioner Essentials, Amazon Web Services (course) [Certificate]

Leadership & Service

- Organizing Secretary, Graduate Student Association of Bangladesh, University of Houston, 2022–2025
- Academic Advisor and Project Mentor, KUET, 2017–2021